Type MS · Finding the unknown letters in a rational curve

Answer 1 of 11 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number		
3 (a)	$c=2,\ d=3$	B1B1 [2]
(b)	$a-bx=0 \Rightarrow x=\frac{a}{b}=\frac{5}{4} \Rightarrow a=5,\ b=4$	M1A1 [2]
(c)	When the curve crosses the y-axis $x=0$ $y=\frac{5-4\times 0}{2\times 0-3}=-\frac{5}{3} \Rightarrow p=-\frac{5}{3}$	M1A1 [2]
(d)	$y=-2$	B1ft [1]
		Total 7 marks

Type MS · Finding the unknown letters in a rational curve

Answer 1 of 11 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 2 of 11 · Question 3 part a · 2 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

3 a	$\frac{a}{-2}=4 \Rightarrow a=-8$	M1 A1 (2)
b	$x=\frac{1}{2}$	B1 (1)
c (i)	$\left(\frac{3}{-"-8"},0\right)\left(=\left(\frac{3}{8},0\right)\right)$	B1 ft
c (ii)	$(0,3)$	B1 (2)
d	Asymptote drawn at $y=4$. The asymptote must either be clearly labelled with its equation or pass through the y axis with 4 clearly labelled.	B1
	Asymptote with equation drawn at $x=\frac{1}{2}$ The asymptote must either be clearly labelled with its equation or pass through the x axis with $\frac{1}{2}$ labelled.	B1 ft
	Correct curve drawn with two branches	B1 ft
	"$\frac{3}{8}$" and "3" labelled on x and y axes respectively	B1 ft (4)
		Total 9 marks

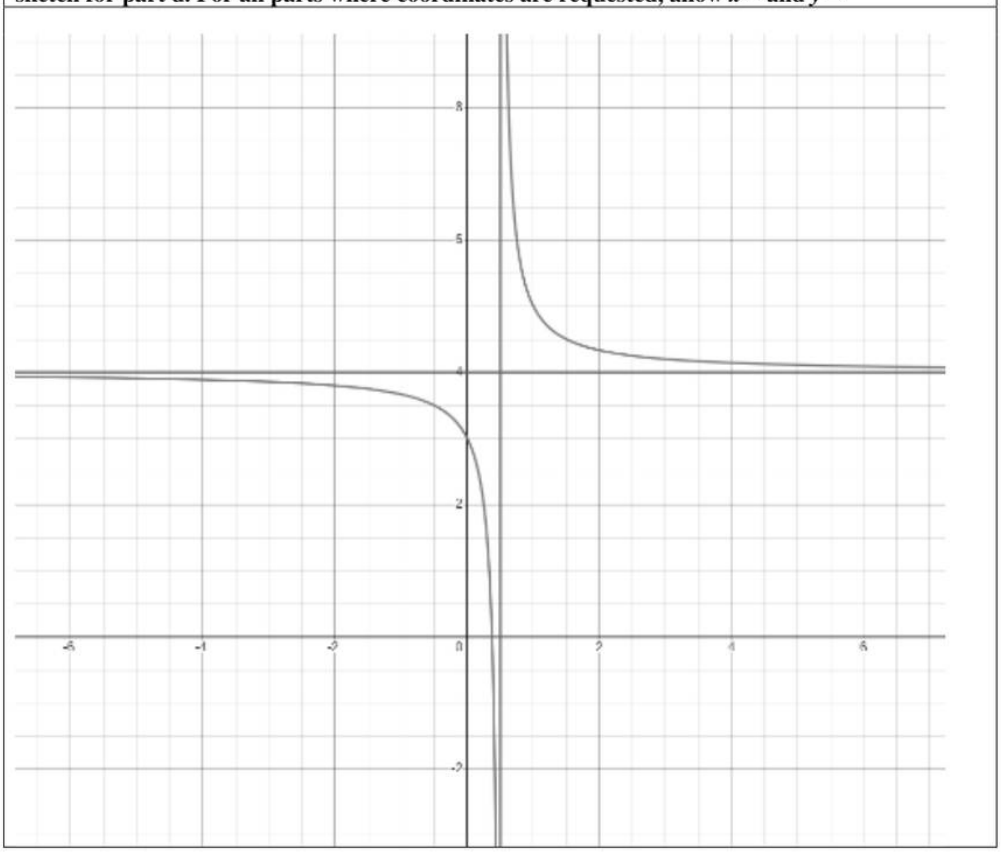
Part	Mark	Additional Guidance
(a)	**M1**	For $\frac{a}{-2}=4$
	A1	For $a=-8$
	Note: it is acceptable if a candidate uses a suitably large value for x and deduces the correct value for a. M1 A1	
(b)	**B1**	For $x=\frac{1}{2}$
(c)(i)	**B1 ft**	For $\left(\frac{3}{-"-8"},0\right)\left(=\left(\frac{3}{8},0\right)\right)$ ft their value from (a). Allow omission of brackets. This mark is awarded for the point at which you see $\left(\frac{3}{-\text{their } a},0\right)$ May be written $x=$
(ii)	**B1**	For $(0,3)$. Allow omission of brackets. May be written $y=$
If there is no labelling of (i) and (ii), it must be unambiguously indicated which coordinate crosses the y axis and which crosses the x axis. If any doubt, do not award the marks.		
(d)	**B1**	For asymptote with equation drawn at $y=4$. The line must be labelled with the equation or must clearly pass through the y-axis labelled as 4. There must be at least one branch of the curve present, which must not cross or bend back from the asymptote.
	B1 ft	For asymptote with equation drawn at $x=\frac{1}{2}$, ft their answer from part b The line must be labelled with the equation or must clearly pass through the x-axis labelled as 0.5. There must be at least one branch of the curve drawn, which must not cross or bend back from the asymptote.

Type MS · Finding the unknown letters in a rational curve

Answer 2 of 11 · Question 3 part a · 2 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

	B1 ft	For correct curve drawn with two branches, in the correct quadrants, ft their answers from part a and b. The curve must be drawn in quadrants 1 and 3.
	B1 ft	For a curve passing through "$\frac{3}{8}$" and "3" labelled on x and y axes respectively, ft their answers from part a, b and c.
Note: answers for parts a, b and c cannot be awarded marks for correct values labelled on the sketch for part d. For all parts where coordinates are requested, allow $x =$ and $y =$		



Type MS · Finding the unknown letters in a rational curve

Answer 2 of 11 · Question 3 part a · 2 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 3 of 11 · Question 2 part whole · 4 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

2	(i)	$\frac{a}{1}=-2$	$a=-2$	B1
	(ii)	$ax+b=0 \Rightarrow -2\times4+b=0$	$b=8$	B1ft
	(iii)	$-3+c=0$	$c=3$	B1
	(iv)	So the equation of S is $y=\frac{8-2x}{x+3}$		
		When $x=0$		
		$y=\frac{8-2\times0}{0+3}=\frac{8}{3}$	$p=\frac{8}{3}$	B1ft [4]
				Total 4 marks

		In parts (i), (ii) and (iii), values must be clearly stated or shown correctly substituted in the equation of the curve.
(ii)	B1ft	ft (i) $b=-4\times$ their a
(iv)	B1ft	ft (ii) and (iii) $p=\frac{\text{their } b}{\text{their } c}$ Accept 2.67 or better, $\left(0,\frac{8}{3}\right)$ or $\frac{8}{3}$ marked on the diagram at the point where the curve crosses the y axis.

Type MS · Finding the unknown letters in a rational curve

Answer 3 of 11 · Question 2 part whole · 4 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 4 of 11 · Question 4 part a.i · 0 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

4(a)(i)	$p = 2$	B1
(ii)	$r = 4$	B1 [2]
(b)	$\frac{3}{2} = \frac{2\times 0+q}{0+'4'} \Rightarrow q = 6$	M1A1 [2]
(c)	$0 = \frac{2s+'6'}{s+'4'} \Rightarrow s = -3$	M1A1 [2]
		Total 6 marks

Question	Notes	Marks
4(a)(i)	$p = 2$	B1 [1]
(ii)	$r = 4$	B1 [1]
(b)	For using the equation for C, substituting $x = 0$ and $y = \frac{3}{2}$ and attempt to rearrange to find the value of q $\frac{3}{2} = \frac{2\times 0+q}{0+4} \Rightarrow q = ...$	M1
	$q = 6$	A1 [2]
(c)	For using the equation of C to find s $0 = \frac{2s+6}{s+4} \Rightarrow s = ...$	M1
	$s = -3$	A1 [2]
		Total 6 marks

Type MS · Finding the unknown letters in a rational curve

Answer 4 of 11 · Question 4 part a.i · 0 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 5 of 11 · Question 4 part a.ii · 2 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

4(a)(i)	$p = 2$	B1
(ii)	$r = 4$	B1 [2]
(b)	$\frac{3}{2} = \frac{2\times 0+q}{0+'4'} \Rightarrow q = 6$	M1A1 [2]
(c)	$0 = \frac{2s+'6'}{s+'4'} \Rightarrow s = -3$	M1A1 [2]
		Total 6 marks

Question	Notes	Marks
4(a)(i)	$p = 2$	B1 [1]
(ii)	$r = 4$	B1 [1]
(b)	For using the equation for C, substituting $x = 0$ and $y = \frac{3}{2}$ and attempt to rearrange to find the value of q $\frac{3}{2} = \frac{2\times 0+q}{0+4} \Rightarrow q = ...$	M1
	$q = 6$	A1 [2]
(c)	For using the equation of C to find s $0 = \frac{2s+6}{s+4} \Rightarrow s = ...$	M1
	$s = -3$	A1 [2]
		Total 6 marks

Type MS · Finding the unknown letters in a rational curve

Answer 5 of 11 · Question 4 part a.ii · 2 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 6 of 11 · Question 4 part b · 2 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

4(a)(i)	$p = 2$	B1
(ii)	$r = 4$	B1 [2]
(b)	$\frac{3}{2} = \frac{2\times 0 + q}{0 + '4'} \Rightarrow q = 6$	M1A1 [2]
(c)	$0 = \frac{2s + '6'}{s + '4'} \Rightarrow s = -3$	M1A1 [2]
		Total 6 marks

Question	Notes	Marks
4(a)(i)	$p = 2$	B1 [1]
(ii)	$r = 4$	B1 [1]
(b)	For using the equation for C, substituting $x = 0$ and $y = \frac{3}{2}$ and attempt to rearrange to find the value of q $\frac{3}{2} = \frac{2\times 0 + q}{0 + 4} \Rightarrow q = ...$	M1
	$q = 6$	A1 [2]
(c)	For using the equation of C to find s $0 = \frac{2s + 6}{s + 4} \Rightarrow s = ...$	M1
	$s = -3$	A1 [2]
		Total 6 marks

Type MS · Finding the unknown letters in a rational curve

Answer 6 of 11 · Question 4 part b · 2 marks

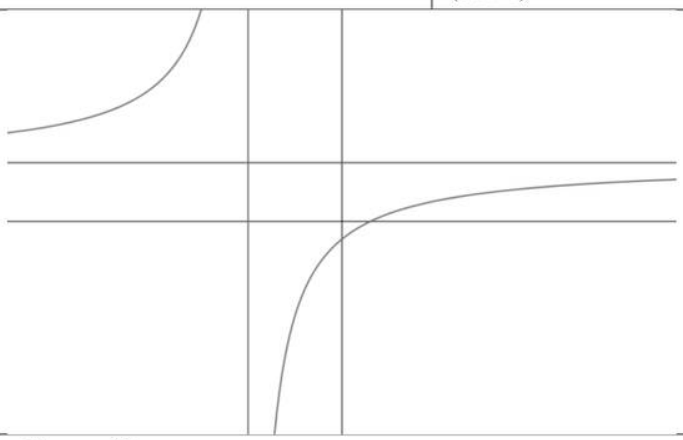
2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 7 of 11 · Question 10 part a.i · 0 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 4 · find part a.i in this mark scheme extract

10(a) Mark (i) and (ii) together	$-\frac{1}{4}=-\frac{2}{b}\rightarrow b=8$ *	B1*cso
	$-\frac{4}{3}a+8=0$ oe or $-\frac{b}{a}=-\frac{4}{3}$	M1
	$a=6$	A1 [3]
(b)	$y=\frac{5}{"6"}$	B1ft [1]
(c)	$\left(\frac{2}{5},0\right)$	B1 [1]
(d)	$\frac{5(6x+8)-6(5x-2)}{(6x+8)^2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion ALT using product rule $(5x-2)(6x+8)^{-1}$ $5(6x+8)^{-1}+(-1)(6)(5x-2)(6x+8)^{-2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion	M1 dM1 A1 A1 [4]
(e)		B1 (curve) B1ft (asymp-totes) B1ft (intersect-ions with axes [3]
(f)	$\frac{13}{100}=\frac{52}{(6x+8)^2}$ oe $(6x+8)^2=400$ **$3x^2+8x-28=0$ oe** $x=2, y=\frac{2}{5}$ $x=-\frac{14}{3}, y=\frac{19}{15}$ oe $\sqrt{\left(("2")-\left("-\frac{14}{3}"\right)\right)^2+\left(\left("\frac{2}{5}"\right)-\left("\frac{19}{15}"\right)\right)^2}$ 6.72	M1 dM1 A1 A1 dM1 A1 [6]
		Total 18 marks

Type MS · Finding the unknown letters in a rational curve

Answer 7 of 11 · Question 10 part a.i · 0 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 4 · find part a.i in this mark scheme extract

Part	Mark	Additional Guidance
(a) (i) and (ii)	B1*cso	For a minimum of the steps shown in the MS to find b, no errors.
	M1	For correct substitution of $x=-\frac{4}{3}$ and $b=8$ into $ax+b=0$
	A1	Correct a.
		Students can complete this in a different order $-\frac{4}{3}a+8=0 \Rightarrow a=6$ $-\frac{b}{a}=-\frac{4}{3}\left(\text{or } -\frac{b}{6}=-\frac{4}{3}\right)\Rightarrow b=8$ B1 M1 A1
(b)	B1ft	Follow through their value of a. $y=\frac{5}{"their 6"}$
(c)	B1	$\left(\frac{2}{5},0\right)$ allow $x=\frac{2}{5}$ $y=0$ $x=\frac{2}{5}$ alone is insufficient
(d)	M1	Attempt the quotient rule. Numerator must be the difference of two terms of the form $A\times(ax+8)-B\times(5x-2)$ or $B\times(5x-2)-A\times(ax+8)$, A and $B>1$. Denominator must be of the form $(ax+8)^2$ $a\neq 0$ Allow follow through of their value of a (general principle of method mark).
	dM1	Either term on the numerator fully correct (either way round), dependent on previous method mark, doesn't need to be simplified. Denominator must be of the form $(ax+8)^2$ Allow follow through of their value of a (general principle of method mark).
	A1	$\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ eg $\frac{13}{(3x+4)^2}$ A simplified expression (denominator expanded acceptable) Dependent on both previous method marks – question states must use calculus, so some method must be shown.
	A1	$\frac{52}{(6x+8)^2}$ and correct conclusion, for example, (the numerator is a positive number) eg the denominator is always positive eg the $(ax+8)^2>0$ and therefore (the fraction) is always positive. No incorrect work for this mark eg don't accept incorrectly simplified gradient.
ALT	M1	Attempt the product rule. $C\times(ax+8)^{-1}+(5x-2)(-(1))(D)\ (ax+8)^{-2}$, C and $D>1$. Allow follow through of their value of a (general principle of method mark).
	M1	Either term correct (either way round), dependent on previous method mark, doesn't need to be simplified
	A1 A1	As main scheme

Type MS · Finding the unknown letters in a rational curve

Answer 7 of 11 · Question 10 part a.i · 0 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 4 · find part a.i in this mark scheme extract

(e)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not <u>obviously</u>** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes $x = -\frac{8}{\text{their } a}$, $y = \frac{5}{\text{their } a}$, ft their value for a, labelled as equations or clearly passing through the correctly labelled numbers on the axis. There must be one section of a negative reciprocal curve present, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through $\left(0, -\frac{1}{4}\right)$ and $\left(\frac{2}{5}, 0\right)$ marked clearly on the graph as coordinates or crossing points on the axis. Ignore any other branch/curve/line present. Allow follow through only of their $(c, 0)$ $c \neq 0$ from part (c)
(f)	M1	Sets their differentiated function from part (d) = $\frac{13}{100}$ Allow any function = $\frac{13}{100}$ coming from working in part (d) or re-started work to find the derivative. The function may not be the original function.
	dM1	For correctly rearranging an equation of the form $\frac{13}{100} = \frac{Q}{(cx+d)^2}$ to $(cx+d)^2 = R$ using their a, or to the form $px^2 + qx + r = 0$ $p, q, r \neq 0$ with 2 out of 3 of p, q, r correct. leading to $x =$
	A1	For $x = 2, y = \frac{2}{5}$ or $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	A1	For $x = 2, y = \frac{2}{5}$ and $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	dM1	For correct use of the formula to find the length of a line, using their values for x and y. Dependent on **first** method mark only.
	A1	For awrt 6.72

Type MS · Finding the unknown letters in a rational curve

Answer 7 of 11 · Question 10 part a.i · 0 marks

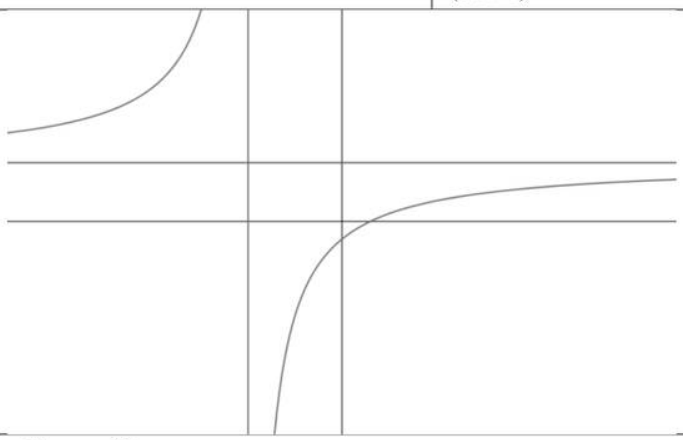
2025 JUN · Paper 02 (regional R) · extract 4 of 4 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 8 of 11 · Question 10 part a.ii · 3 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 4 · find part a.ii in this mark scheme extract

10(a) Mark (i) and (ii) together	$-\frac{1}{4}=-\frac{2}{b}\rightarrow b=8$ *	B1*cso
	$-\frac{4}{3}a+8=0$ oe or $-\frac{b}{a}=-\frac{4}{3}$	M1
	$a=6$	A1 [3]
(b)	$y=\frac{5}{"6"}$	B1ft [1]
(c)	$\left(\frac{2}{5},0\right)$	B1 [1]
(d)	$\frac{5(6x+8)-6(5x-2)}{(6x+8)^2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion ALT using product rule $(5x-2)(6x+8)^{-1}$ $5(6x+8)^{-1}+(-1)(6)(5x-2)(6x+8)^{-2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion	M1 dM1 A1 A1 [4]
(e)		B1 (curve) B1ft (asymptotes) B1ft (intersections with axes [3]
(f)	$\frac{13}{100}=\frac{52}{(6x+8)^2}$ oe	M1
	$(6x+8)^2=400$ **$3x^2+8x-28=0$ oe**	dM1
	$x=2, y=\frac{2}{5}$ $x=-\frac{14}{3}, y=\frac{19}{15}$ oe	A1 A1
	$\sqrt{\left(("2")-\left("-\frac{14}{3}"\right)\right)^2+\left(\left("\frac{2}{5}"\right)-\left("\frac{19}{15}"\right)\right)^2}$	dM1
	6.72	A1 [6]
		Total 18 marks

Type MS · Finding the unknown letters in a rational curve

Answer 8 of 11 · Question 10 part a.ii · 3 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 4 · find part a.ii in this mark scheme extract

Part	Mark	Additional Guidance
(a) (i) and (ii)	B1*cso	For a minimum of the steps shown in the MS to find b, no errors.
	M1	For correct substitution of $x=-\frac{4}{3}$ and $b=8$ into $ax+b=0$
	A1	Correct a.
		Students can complete this in a different order $-\frac{4}{3}a+8=0 \Rightarrow a=6$ $-\frac{b}{a}=-\frac{4}{3}\left(\text{or } -\frac{b}{6}=-\frac{4}{3}\right)\Rightarrow b=8$ B1 M1 A1
(b)	B1ft	Follow through their value of a. $y=\frac{5}{"their 6"}$
(c)	B1	$\left(\frac{2}{5},0\right)$ allow $x=\frac{2}{5}$ $y=0$ $x=\frac{2}{5}$ alone is insufficient
(d)	M1	Attempt the quotient rule. Numerator must be the difference of two terms of the form $A\times(ax+8)-B\times(5x-2)$ or $B\times(5x-2)-A\times(ax+8)$, A and $B>1$. Denominator must be of the form $(ax+8)^2$ $a\neq 0$ Allow follow through of their value of a (general principle of method mark).
	dM1	Either term on the numerator fully correct (either way round), dependent on previous method mark, doesn't need to be simplified. Denominator must be of the form $(ax+8)^2$ Allow follow through of their value of a (general principle of method mark).
	A1	$\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ eg $\frac{13}{(3x+4)^2}$ A simplified expression (denominator expanded acceptable) Dependent on both previous method marks – question states must use calculus, so some method must be shown.
	A1	$\frac{52}{(6x+8)^2}$ and correct conclusion, for example, (the numerator is a positive number) eg the denominator is always positive eg the $(ax+8)^2>0$ and therefore (the fraction) is always positive. No incorrect work for this mark eg don't accept incorrectly simplified gradient.
ALT	M1	Attempt the product rule. $C\times(ax+8)^{-1}+(5x-2)(-(1))(D)\ (ax+8)^{-2}$, C and $D>1$. Allow follow through of their value of a (general principle of method mark).
	M1	Either term correct (either way round), dependent on previous method mark, doesn't need to be simplified
	A1 A1	As main scheme

Type MS · Finding the unknown letters in a rational curve

Answer 8 of 11 · Question 10 part a.ii · 3 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 4 · find part a.ii in this mark scheme extract

(e)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes $x = -\frac{8}{\text{their } a}$, $y = \frac{5}{\text{their } a}$, ft their value for a, labelled as equations or clearly passing through the correctly labelled numbers on the axis. There must be one section of a negative reciprocal curve present, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through $\left(0, -\frac{1}{4}\right)$ and $\left(\frac{2}{5}, 0\right)$ marked clearly on the graph as coordinates or crossing points on the axis. Ignore any other branch/curve/line present. Allow follow through only of their $(c, 0)$ $c \neq 0$ from part (c)
(f)	M1	Sets their differentiated function from part (d) = $\frac{13}{100}$ Allow any function = $\frac{13}{100}$ coming from working in part (d) or re-started work to find the derivative. The function may not be the original function.
	dM1	For correctly rearranging an equation of the form $\frac{13}{100} = \frac{Q}{(cx+d)^2}$ to $(cx+d)^2 = R$ using their a, or to the form $px^2 + qx + r = 0$ $p, q, r \neq 0$ with 2 out of 3 of p, q, r correct. leading to $x =$
	A1	For $x = 2, y = \frac{2}{5}$ or $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	A1	For $x = 2, y = \frac{2}{5}$ and $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	dM1	For correct use of the formula to find the length of a line, using their values for x and y. Dependent on **first** method mark only.
	A1	For awrt 6.72

Type MS · Finding the unknown letters in a rational curve

Answer 8 of 11 · Question 10 part a.ii · 3 marks

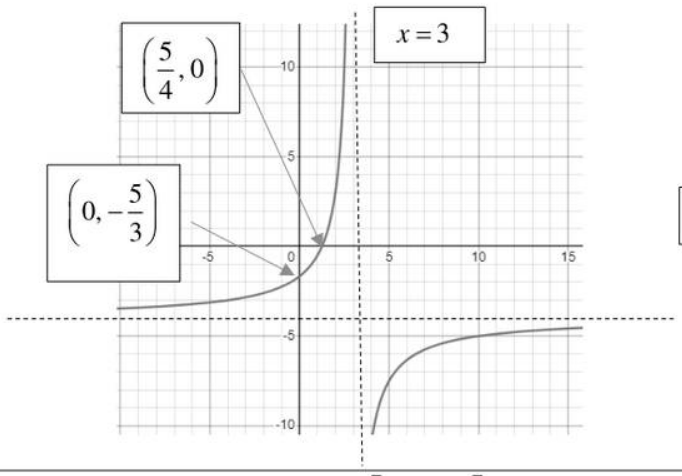
2025 JUN · Paper 02 (regional R) · extract 4 of 4 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 9 of 11 · Question 10 part a · 2 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

10(a)	$\left(ax-5=0,\ x=\frac{5}{4}\Rightarrow\right)a=4$ $\left(\text{Line parallel to the } y\text{-axis is } x=3\Rightarrow\right)b=3$	B1 B1 {2}
(b)		B1 B1ftB1 B1ftB1ft [5]
(c)	$\left(\frac{dy}{dx}=\right)\frac{(3-x)(4)-(4x-5)(-1)}{(3-x)^2}=\left[\frac{7}{(3-x)^2}\right]$ (Gradient of l =) $\frac{7}{4}$ $\frac{7}{4}=\frac{7}{(3-x)^2}\Rightarrow(3-x)^2=4\Rightarrow 3-x=\pm 2\Rightarrow x=1,5$ When $x=1$, $y=-\frac{1}{2}$ and when $x=5$, $y=-\frac{15}{2}$ Equation of line when $x=1$: $y--\frac{1}{2}=\frac{7}{4}(x-1)\Rightarrow 4y-7x=-9$ Equation of line when $x=5$: $y--\frac{15}{2}=\frac{7}{4}(x-5)\Rightarrow 4y-7x=-65$ $\Rightarrow -65<k<-9$	M1 B1 M1A1 M1 A1 (B1B1 on ePen) ddM1 ddM1 A1 [9]

Type MS · Finding the unknown letters in a rational curve

Answer 9 of 11 · Question 10 part a · 2 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

ALT	$(y=)\dfrac{7x+k}{4}=\dfrac{"4"x-5}{"3"-x}$ oe $4\left(\dfrac{"4"x-5}{"3"-x}\right)-7x=k$ oe	M1 A1(B1 on ePen)
	$16x-20=(7x+k)("3"-x)$ oe	
	$16x-20=21x-7x^2+3k-kx \Rightarrow 7x^2+kx-5x-20-3k(=0)$ oe	M1 A1
	$(\Rightarrow 7x^2+(k-5)x-20-3k(=0))$ oe	
	$(k-5)^2-4(7)(-20-3k)(<0)$	M1A1 (B1 B1 on ePen)
	$k^2+74k+585(<0)$	
	$(k=)\dfrac{-74\pm\sqrt{(74)^2-4\times(1)\times585}}{2}$ or $(k-9)(k-65)(=0)$	ddM1 ddM1
	$-65<k<-9$	A1 [9]

Type MS · Finding the unknown letters in a rational curve

Answer 9 of 11 · Question 10 part a · 2 marks

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Part	Mark	Notes
(a)	B1	For the value of $a = 4$ or $b = 3$
	B1	For the value of $a = 4$ and $b = 3$
Note: These are independent of method marks, not given answers. Unless the answers come from **very obviously** incorrect working, these marks should be awarded for sight of the correct values.		
(b)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn and **must not** **obviously** 'bend back' on themselves. Mark intention.
	For the following marks, where candidates have used incorrect values of a and b, examiners will find using desmos.com or similar packages useful. Type into desmos $y = (ax - b)/(b - x)$ and add 'sliders' for a and b	
	B1ft	For the horizontal asymptote of $y = -4$ drawn in the correct place and labelled with its equation or where the line passes through the axis. There **must** be at least one branch of a reciprocal curve present which **must** not cross or **obviously** bend back from the asymptote(s) ignore any other curves present. Allow follow through **only** of $y = -$ (their a) and only if other conditions met.
	B1	For the vertical asymptote $x = 3$ drawn in the correct place. There **must** be at least one branch of a reciprocal curve present which **must** not cross or **obviously** bend back from the asymptote(s), ignore any other curves present.
	B1ft	A single curve (or even line) passing through $\left(0, -\frac{5}{3}\right)$ marked clearly on the graph as a coordinate or as a crossing point on the axis. Ignore any other branches present. Allow follow through only of $\left(0, -\frac{5}{\text{their } b}\right)$
	B1ft	A single curve (or even line) passing through both $\left(\frac{5}{4}, 0\right)$ and $\left(0, -\frac{5}{3}\right)$ both marked clearly on the graph as a coordinate or as crossing points on the axes. Ignore any other branches present. Allow follow through only of $\left(0, -\frac{5}{\text{their } b}\right)$

Type MS · Finding the unknown letters in a rational curve

Answer 9 of 11 · Question 10 part a · 2 marks

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(c)	M1	For applying the quotient (or product rule). • With two terms in the numerator subtracted either way around. • The differentiation on both terms must be correct. • The denominator must be squared. $\frac{(3-x)(4)-(4x-5)(-1)}{(3-x)^2}$ or $\frac{(4x-5)(-1)-(3-x)(4)}{(3-x)^2}$ (if using quotient rule) Or $4(3-x)^{-1} \pm (4x-5)(3-x)^{-2}$ (if using product rule)
	B1	For (Gradient of l =) $\frac{7}{4}$
	M1	For correctly setting their gradient of l = to their derivative and an attempt to solve to find 2 values of x. Allow one error in rearrangement. Their derivative must be of the form $\frac{rx+s}{(3-x)^2}$ s **or** r could be 0, but not both
	A1	For $x = 1$ **and** 5
	M1 (B1 on ePen)	For substitution of either of their x values into the equation of their curve.
	A1 (B1 on ePen)	For $y = -\frac{1}{2}$ **and** $y = -\frac{15}{2}$
	ddM1	For correctly forming an equation of the line for either pair of their values and their gradient and rearranging (allow errors in rearrangement) to the form $4y - 7x = k$. Dependent on 1st 2 method marks.
	ddM1	For correctly forming an equation of the line for both pairs of their values and their gradient and rearranging (allow rearrangement errors) to the form $4y - 7x = k$ Dependent on 1st 2 method marks.
	A1	For the correct inequality as shown.
ALT	M1	For rearrangement of l to the form $y = \frac{7}{4}x + c \quad c \neq 0$ and placing equal to their equation of C or substitution of the equation for curve C in the equation for l. Allow $4\left(\frac{"4"x-5}{"3"-x}\right) = k \pm 7x$ Condone using an inequality sign, follow through their a and b
	A1 (B1 on ePen)	For a correct unsimplified equation. **DO NOT** condone using an inequality sign
	M1	For an attempt to rearrange their equation of the form $mx + c = \frac{4x-5}{3-x}$ $m, c \neq 0$, to allow them to reach a "3TQ", allow one error in rearrangement. Condone using an inequality sign By "3TQ", we means terms in x, x^2 and constant, all on 'one side' even if the equals sign is missing or an incorrect inequality sign is shown, terms simplified where possible.
	A1	For the correct "3TQ" shown (oe), the term in x does not have to be factorised and = 0 may be omitted. "3TQ" defined as above
	M1 (B1 on ePen)	For the correct formulation of the discriminant from their "3TQ". Inequality sign doesn't need to be shown or can be incorrect.
	A1 (B1 on ePen)	For the correct unsimplified discriminant. Inequality sign doesn't need to be shown or can be correct.
	ddM1	For a correct method to solve their quadratic equation in k, leading to a value of k. Dependent on first 2 method marks. Mark awarded for the correct **method.**
	ddM1	For a correct method to solve their quadratic equation in k, leading to two values of k. Dependent on 1st 2 method marks. Mark awarded for the correct **method.**
	A1	For the inequality shown.

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Answer 9 of 11 · Question 10 part a · 2 marks

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Answer 10 of 11 · Question 3 part a · 2 marks

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Number		
3 (a)	$c=2,\ d=3$	B1B1 [2]
(b)	$a-bx=0 \Rightarrow x=\frac{a}{b}=\frac{5}{4} \Rightarrow a=5,\ b=4$	M1A1 [2]
(c)	When the curve crosses the y-axis $x=0$ $y=\frac{5-4\times 0}{2\times 0-3}=-\frac{5}{3} \Rightarrow p=-\frac{5}{3}$	M1A1 [2]
(d)	$y=-2$	B1ft [1]
		Total 7 marks

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Answer 10 of 11 · Question 3 part a · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the unknown letters in a rational curve

Answer 11 of 11 · Question 11 part b.ii · 1 mark

2022 JUN · Paper 02 · extract 1 of 4 · find part b.ii in this mark scheme extract

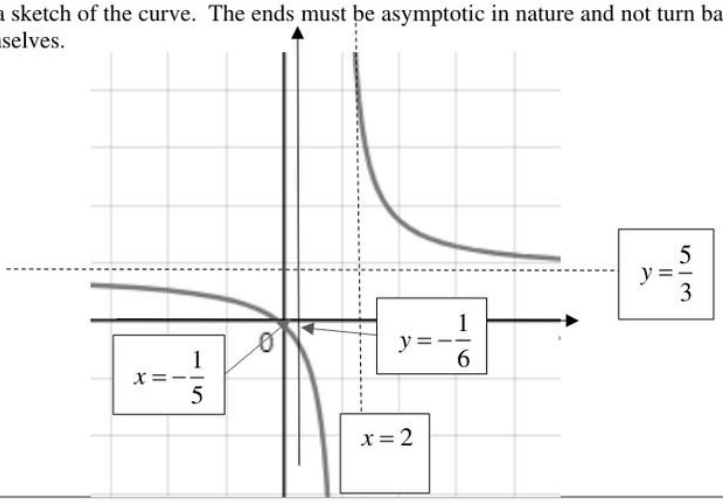
11 a	$\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$	M1A1A1 (3)
b (i)	At $x = 0$ $\frac{dy}{dx} = -\frac{11}{12}$	B1
	So $\frac{dy}{dx} = \frac{-6(2a-1)-a}{36} = -\frac{11}{12}$	M1
	$-13a+6 = -33 \Rightarrow a = 3$*	dM1 A1 cso (4)
b (ii)	$y = \frac{5x+1}{3x-6}$*	B1 cso (1)
c	Asymptote with equation drawn at $y = 2$	B1
	Asymptote with equation drawn at $x = \frac{5}{3}$	B1
	Correct curve drawn with two branches	B1
	$-\frac{1}{5}$ and $-\frac{1}{6}$ labelled on x and y axes	B1 B1 (5)
d	$\frac{12}{11}x - \frac{1}{6} = \frac{5x+1}{3x-6}$	M1
	$(3x-6)(72x-11) = 66(5x+1)$	M1
	$x(72x-265) = 0$ oe	M1
	$x = \frac{265}{72}$ oe	A1 (4)
		Total 17 marks

Part	Mark	Notes
(a)	**M1**	For attempting to use a correct quotient rule: The definition of an attempt is as follows: • The denominator must be correct • The terms in the numerator must be subtracted either way around • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $ax-6 \to \pm a$
	A1	For one term in the numerator correct
	A1	For a fully correct differentiated expression. $\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$ No simplification is required.

Type MS · Finding the unknown letters in a rational curve

Answer 11 of 11 · Question 11 part b.ii · 1 mark

2022 JUN · Paper 02 · extract 2 of 4 · find part b.ii in this mark scheme extract

	Uses Product Rule	
	M1	For attempting to use a correct product rule: The definition of an attempt is as follows: • The terms in the numerator must be added. • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $(ax-6)^{-1} \to \pm a(ax-6)^{-2}$
	A1	For one term correct
	A1	For a fully correct differentiation $\frac{dy}{dx} = \left[(2a-1)x+1\right](a)(ax-6)^{-2} + (ax-6)^{-1}(2a-1)$
(b)(i)	**B1**	For finding the gradient of the tangent $= -\frac{11}{12}$
	M1	For using that at the point A $x = 0$ and setting their $\frac{dy}{dx} = \left(-\frac{11}{12}\right)$ which must be the gradient of their tangent to obtain $\frac{-6(2a-1)-a}{36} = '-\frac{11}{12}' \Rightarrow \left(\frac{-13a+6}{36} = -\frac{11}{12}\right)$
	dM1	For simplifying to a linear equation and attempting to solve.
	A1	For $a = 3$ with no errors seen
(b)(ii)	**B1**	For $y = \frac{5x+1}{3x-6}$ exactly in that form. You must see evidence of substitution of $a = 3$ for this mark. There must be $y =$ …. [Condone missing $x \neq 2$]
(c)		This is a sketch of the curve. The ends must be asymptotic in nature and not turn back on themselves. 
	B1	For asymptote **with equation drawn** at $y = \frac{5}{3}$ OR $x = 2$
	B1	For both asymptotes **with equations drawn** at $y = \frac{5}{3}$ AND $x = 2$
	B1	Correct curve drawn with two branches with one branch in 1st Quadrant and another branch in Quadrants 2, 3 and 4 with the graph drawn the correct way around. The graph must be asymptotic in nature.

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Answer 11 of 11 · Question 11 part b.ii · 1 mark

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	B1	$-\frac{1}{5}$ labelled on x-axis **or** $-\frac{1}{6}$ labelled on y- axis
	B1	$-\frac{1}{5}$ labelled on x axis **and** $-\frac{1}{6}$ labelled on y- axis
(d)	**M1**	For setting $\frac{12}{11}x-\frac{1}{6}=\frac{5x+1}{3x-6}$ [eliminating y from both equations]
	M1	For forming a quadratic equation, 2TQ or 3TQ in any order. $(3x-6)(72x-11)=66(5x+1) \Rightarrow 72x^2-265x=0$
	M1	For solving their QE provided it is either a 2TQ or a 3TQ
	A1	For $x=\frac{265}{72}$ oe

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Answer 11 of 11 · Question 11 part b.ii · 1 mark

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